

Defense-Related R&D and the Growth of the Postwar Information Technology Industrial Complex in the United States

Kira R. Fabrizio, David C. Mowery

Citer ce document / Cite this document :

Fabrizio Kira R., Mowery David C. Defense-Related R&D and the Growth of the Postwar Information Technology Industrial Complex in the United States. In: Revue d'économie industrielle, vol. 112, 4e trimestre 2005. Réseaux de connaissances, réseaux de défense. pp. 27-44;

doi : <https://doi.org/10.3406/rei.2005.3123>;

https://www.persee.fr/doc/rei_0154-3229_2005_num_112_1_3123;

Fichier pdf généré le 17/04/2024

Abstract

The U.S federal government, especially the Department of Defense, played a substantial role in the development of the electronic computer, computer software, and semiconductor components industries, and the resulting development of the Internet. Federal R&D investment, procurement policies, and spillovers of technologies from military to civilian applications contributed to the industrial and technological development of these sectors. We examine the changing role of the federal government in these sectors over time, highlighting the critical importance of such involvement in early industry development and the fading of this role over time. We conclude with some discussion of the potential implications of post-9/11 changes in federal R&D spending.

Résumé

Le gouvernement fédéral américain, plus particulièrement le ministère de la Défense, a joué un rôle important dans le développement des ordinateurs, des logiciels et des semi-conducteurs, et dans les développements induits de l'Internet. Les investissements de R&D du gouvernement fédéral, les spillovers technologiques des applications militaires vers le civil ont contribué aux développements industriels et technologiques de ces secteurs. Les auteurs analysent les évolutions du rôle du gouvernement fédéral dans ces secteurs dans le temps, en mettant en évidence l'importance fondamentale de son implication dans le développement naissant de l'industrie et l'effacement progressif de ce rôle dans le temps. Ils concluent par une discussion des implications potentielles des changements post-9/11 dans les dépenses de R&D du gouvernement fédéral.

Kira R. FABRIZIO
Goizueta Business School - Emory University

David C. MOWERY
Haas School of Business - U.C. Berkeley

DEFENSE-RELATED R&D AND THE GROWTH OF THE POSTWAR INFORMATION TECHNOLOGY INDUSTRIAL COMPLEX IN THE UNITED STATES

Mots-clés : Financement public, technologies de l'information, secteur émergent

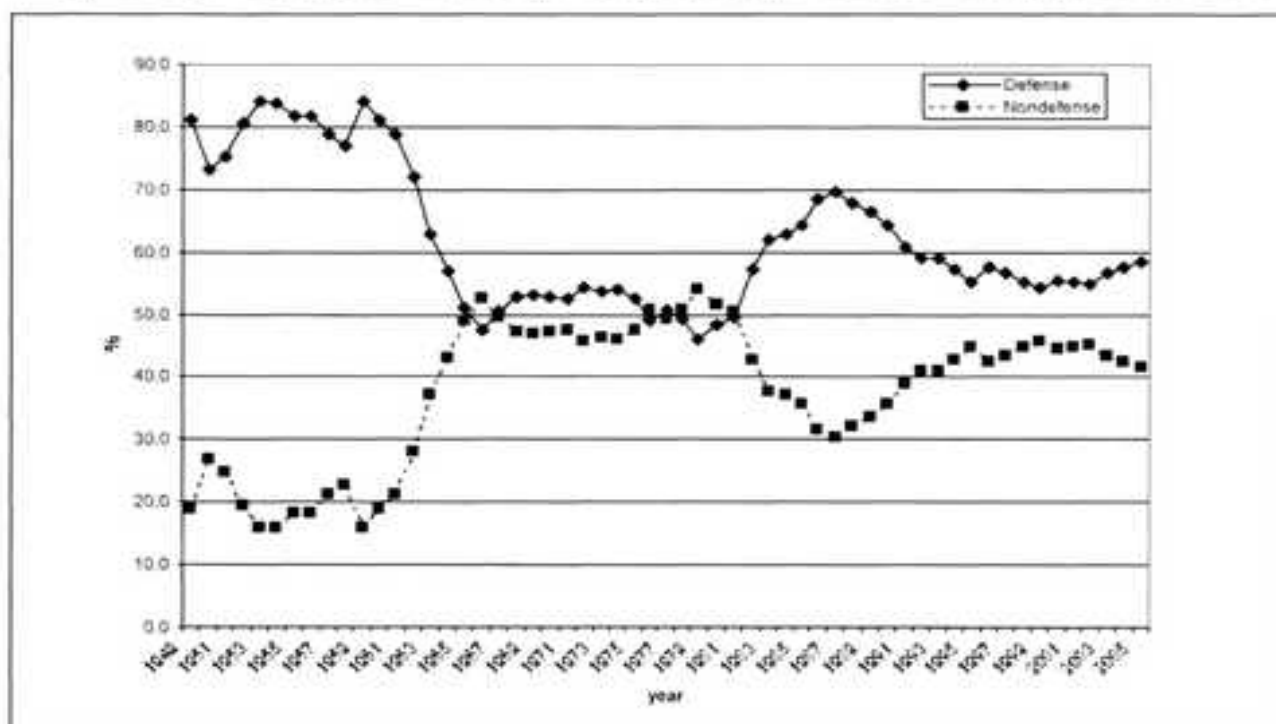
Key words : Federal Funding, IF sector, Industry Development

INTRODUCTION

No summary of major postwar innovations in U.S. economic history can ignore the information technology (IT) sector. Advances in electronics and related technologies created three new industries – electronic computers, computer software, and semiconductor components, and these three industries combined to give birth to the Internet, a « general purpose technology » spanning many industrial sectors. The technologies underpinning the industries that we include in our definition of the IT sector, as well as the development of

(*) This paper draws on a presentation at the SSRC conference on « Financing Major Innovations », Irvine, California, March 22, 2003, and benefited from comments from the participants at that meeting, especially Linda Cohen, Steve Usselman and Zhorina Khan. Research for this paper was supported by the Andrew W. Mellon Foundation and the Kauffman Foundation.

Figure 1
Defense & Nondefense share of total federal gov't R&D outlays, 1949-2005

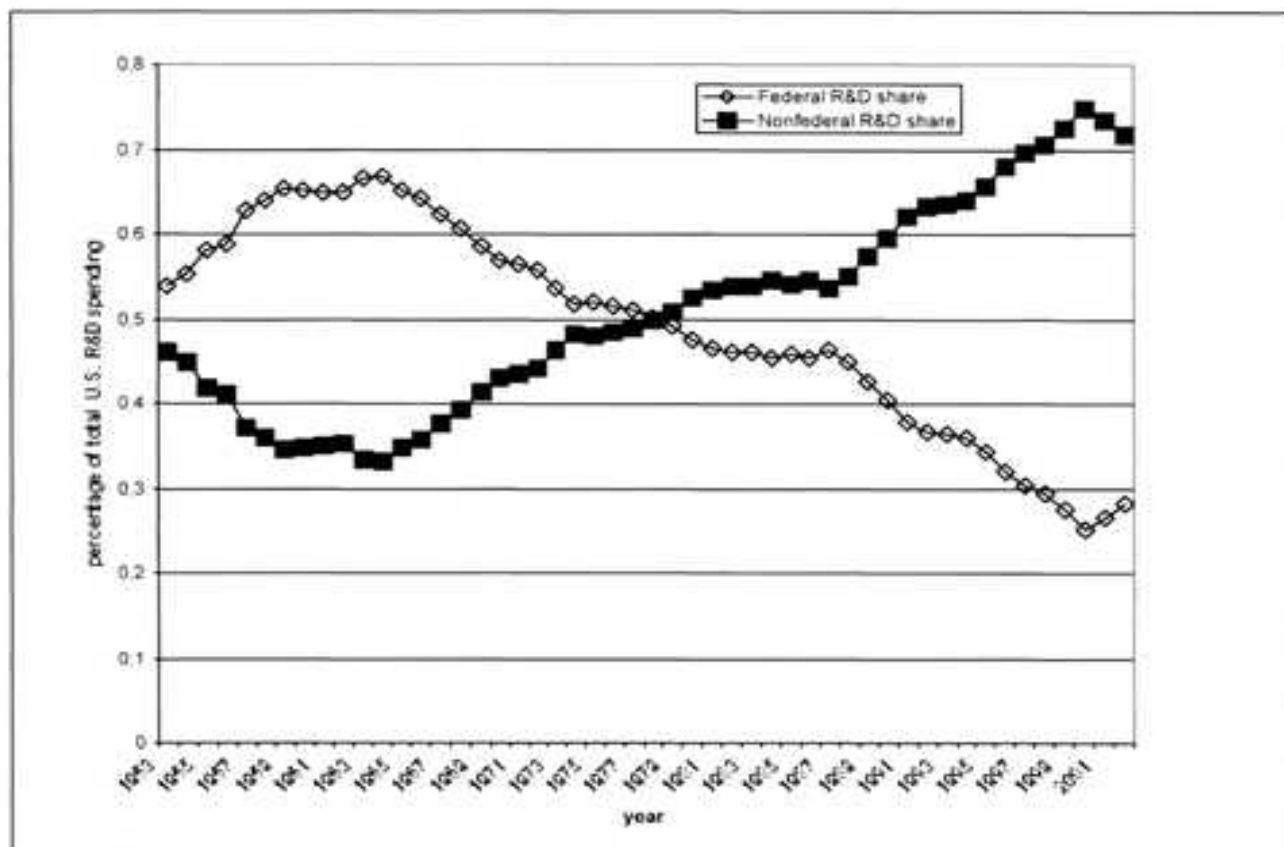


these industries, were heavily influenced by the relationships among the federal government, academia, and industry.

It is impossible to overlook the role of the federal government, especially the U.S. Department of Defense (DoD), in financing innovation in these technologies. Indeed, the Defense Department accounted for a substantial share of federal and overall U.S. R&D spending for much of the postwar period. Defense-related R&D spending accounted for more than 80 % of total federal R&D spending for much of the 1950s, and rarely has dropped below 50 % of federal R&D expenditures during the entire 1949-2005 period (Figure 1). Since federal R&D spending accounted for more than 50 % of total national R&D spending during 1953-78 (data for national R&D investment are available only after 1952), and only dropped below 40 % after 1990 (Figure 2), the significance of the federal government's defense-related R&D investment is obvious—in some years during the postwar period (*e.g.*, the late 1950s and early 1960s), defense-related R&D investment accounted for nearly one-half of total national R&D spending.

The U.S. defense R&D budget, like that of most other industrial economies, is dominated by development (« D ») spending, which typically accounts for 85-90 % of annual federal spending on defense-related R&D. Defense-related spending on « basic » and « applied » research together accounted for only slightly more than 10 % of overall federal spending on these activities during fiscal 2001 (the last budget of the Clinton Administration, and one that was not

Figure 2 - Federal and Nonfederally funded R&D, 1953-2002



revised in response to the 9/11 attacks). Defense-related « R » investments (defining these to include basic and applied research activities as defined by the U.S. Defense Department) nevertheless account for a significant share of overall federal support for research in such fields as computer science (35 % in fiscal 2001) or engineering (more than 30 %; all figures from American Association for the Advancement of Science, 2002). In fields such as aeronautics, electrical engineering, computer science, and oceanography, defense-related research spending has made significant contributions to the creation of a university-based « research infrastructure » during the postwar period that has been an important source of civilian innovations, new firms, and trained scientists and engineers. Indeed, defense-related R&D spending was a major force in the creation of what some have referred to as the « Cold War University », notably at such institutions as Stanford University and MIT, both of which relied on federal research support in basic and applied fields of science and engineering to expand and strengthen their academic eminence (Lowen, 1997; Leslie, 1993).

The economic and technological effects of the enormous postwar U.S. investment in defense-related R&D remain controversial. Indeed, interpretations of these effects are highly correlated with changing views of the economic competitiveness of the United States. During the 1980s and early 1990s, for example, defense-related investments in R&D and procurement often were

criticized for spawning small, financially weak firms in such industries as semiconductors (1) and in the information technology sector more generally. Another line of criticism of the economic benefits from defense-related R&D spending argued that this form of R&D investment could « crowd out » privately financed R&D on purely civilian technologies (See Cowan and Foray, 1995), and several empirical studies of overall defense-related R&D investment and economic returns found very low rates of social return (Lichtenberg, 1984; Schankerman and Pakes, 1986). Still other critics argued that a reversal in the direction of military – civil technology spillovers was degrading the performance of military systems, because of the asserted failure of defense contractors to adopt advances in both product and process technologies from civilian applications (Samuels, 1994).

Since the mid-1990s, much of this criticism of the burdens of defense-related R&D has been muted in the face of the advances in information technology that appear to have contributed to the productivity surge associated with the « New Economy » of the late 1990s. A more disaggregated evaluation of the effects of defense-related R&D yields a mixed verdict, one that is further confused by changing assessments of overall U.S. economic performance.

This paper reviews the history of defense-related R&D and procurement support for the development of the key industries in the IT sector. The IT sector, which scarcely existed in 1945, was a key focus of DoD R&D and procurement spending for much of the postwar period. These R&D and procurement programs affected both the development of the underlying technologies and the structure of the industries that developed these technologies for defense and civilian applications. Indeed, the structure of the U.S. semiconductor, computer hardware, and computer software industries all differ in varying degrees from those of their counterparts in other industrial economies. The scale and structure of U.S. defense-related R&D and procurement programs are responsible in part for these structural contrasts.

There are at least three primary channels through which defense-related public investments affect national innovative performance. Defense-related

- (1) The unusual structure of the U.S. semiconductor manufacturing industry, dominated in many segments by vertically specialized, « merchant » firms, was criticized as one that weakened leading firms, which lacked the large financial reserves of their large Japanese and European competitors (See Best, 1996; Borrus, 1988; Dertouzos *et al.*, 1989). The waves of entry by new firms into semiconductor production also were criticized, for producing financially weak firms that failed to focus on developing strong manufacturing capabilities (See Florida and Kenney, 1990). Many of the characteristics of the U.S. semiconductor industry that were the focus of these criticisms during the 1980s and early 1990s were celebrated as sources of competitive strength during the late 1990s, suggesting that either the critique or the celebration of U.S. performance require further evaluation.

R&D investments may support the creation of new bodies of scientific or engineering knowledge that support innovation in both defense-related and civilian applications. Such investments also influence the development of important institutional components of national innovation systems, such as university-based research, that provide both research and trained scientists and engineers. This channel of interaction is likely to be most responsive to defense-related investments in basic and applied research, rather than development.

A second important channel through which defense-related R&D investments affect civilian innovative performance are the classic « spinoffs », where defense-related R&D programs yield technologies that have important applications in civilian products. This channel of interaction can benefit from defense-related investments in technology development, as well as research. But the civilian « spinoffs » associated with defense-related investments in « D » appear to be most significant in the early development of new technologies, since these early phases are more likely to be characterized by substantial overlap between defense and nondefense requirements for applications. As technologies mature (often associated with declining unit costs and widening applications in civilian and military sectors), civilian and military requirements frequently diverge, and the scope for civilian benefit from such « spinoffs » declines.

A third important channel through which defense-related spending on new technologies often benefits civilian applications is procurement. Defense-related R&D investments frequently lead to substantial government purchases of new technologies. The military services, whose requirements often emphasize performance above all other characteristics (including cost), have on a number of occasions functioned as a « lead purchaser », placing large orders for early versions of new technologies. These procurement programs in many cases enable supplier firms to reduce the costs of their products through learning effects. In addition, military applications may provide a basis for improving the reliability and functionality of such technologies. New technologies undergo a prolonged period of « debugging », performance and reliability improvement, cost reduction, and learning on the part of users and producers about applications and maintenance (Mowery and Rosenberg, 1999). The pace and pattern of such progressive improvement affect the rate of adoption, and the rate of adoption in turn affects the development of these innovations. Government procurement allowed companies in these industries to benefit from learning by increasing the scale of production for early stages of the technology.

As we show in our discussion below, the development of the U.S. IT sector gained from all three of these sources of benefits. Nevertheless, the effects of defense-related R&D and procurement investments on civilian innovation and applications development in IT appear to have weakened during the 5-decade period discussed in this paper. This tendency for defense-related investments to yield more modest civilian innovative payoffs could be exacerbated by

shifts in the allocation of U.S. defense-related R&D and procurement investments in the wake of the 9/11 catastrophe, as we note in our conclusion.

DoD FUNDING AND IT INNOVATION

Federal policy played a central role in the development of all four of the technologies that we include in our definition of the IT sector. In all of these technologies, the direct influence of DoD R&D and procurement policies was strongest in the early years of their development, when defense-related expenditures on R&D and/or procurement accounted for the largest share of such funding. In the early stages of these technologies, the DoD funded basic research, provided funds for investment and development, and generated sufficient demand to support cost-reducing learning and production scale economies that broadened the market for products based on these technologies.

The semiconductor, computer hardware and software, and Internet « industries » now encompass many markets and applications beyond national defense, which account for a much smaller share of demand or applications in all of these industries. Indeed, the technological « spillovers » that once flowed from defense-related technologies to civil applications now frequently move in the opposite direction, and the ability of Defense Department policymakers to influence the direction of technological change has diminished considerably. Nonetheless, the substantial presence of DoD support programs in these industries' early years contributed to the development of intellectual property policies, patterns of entry, and overall industrial structure that remain very distinctive in the IT sector of the 21st century.

The electronics revolution that spawned the semiconductor and computer industries, as well as the Internet, can be traced to two key innovations – the transistor and the computer. Both appeared in the late 1940s, and the exploitation of both was spurred by Cold War concerns over national security. The creation of these innovations also relied on domestic U.S. science and invention to a greater extent than many of the critical innovations of the pre-1940 era. The following sections briefly survey the development of each of these technological advances, describing key aspects of their industrial and technological evolution and highlighting the role of the federal government in financing major innovations and supporting industrial development.

Semiconductors

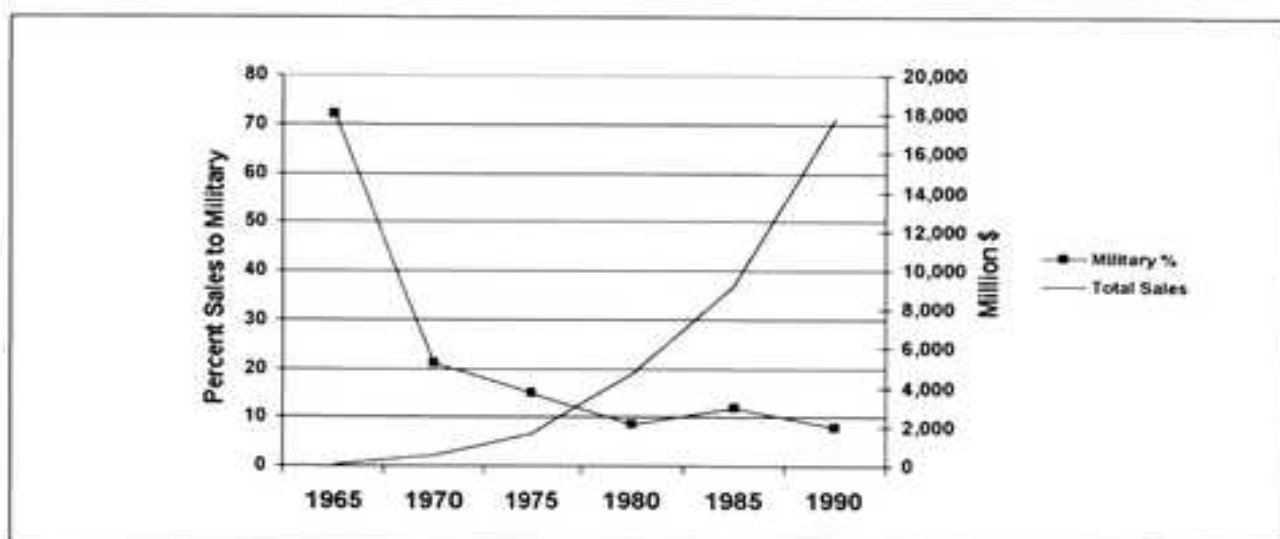
The transistor was invented at Bell Telephone Laboratories in late 1947, and marked one of the first payoffs to the basic research program in solid-state physics that Mervin Kelly, Bell Labs' director, had launched in the 1930s. The transistor had important potential applications in military electronics and computer systems, and federal funds, largely from DoD, the Atomic Energy Commission, or other defense-related agencies, accounted for nearly 25 % of

total industry R&D spending in the late 1950s. The bulk of this defense-related R&D spending during the 1950s was allocated to established producers of electronic components, who were not among the pioneers in the introduction of innovations in semiconductor technology. Paradoxically, the firms responsible for many of the key early innovations in semiconductors did so without military R&D contracts, relying instead on support from procurement contracts (Kleiman, 1966, pp. 173-174). As a result, the R&D facilities of large U.S. firms provided many of the basic technological advances that were commercialized by new entrants.

The silicon junction transistor, the first transistor sufficiently robust for deployment in military systems, was produced by Texas Instruments, rather than AT&T, in 1954. The next major advance in semiconductor electronics was the integrated circuit (IC). The IC was largely invented in 1958 by Jack Kilby of Texas Instruments, who sought to develop a device that would meet the military's demand for more reliable semiconductor components; little of Kilby's pathbreaking R&D was supported by the U.S. military. Although government purchases of integrated circuits, which were largely devoted to military applications, dominated the early years of IC production, the federal government share of total IC demand declined after the mid-1960s (Figure 3).

Although it rapidly shrank as a share of total demand for ICs (by the 1990s, the U.S. military accounted for less than 10 % of IC sales), military demand provided opportunities for learning-by-doing and expanded production that in turn spurred output growth and price reductions that expanded commercial demand for ICs. Military procurement policies also influenced industry structure by promoting competition and the intra-industry diffusion of technological knowledge. In contrast to Western European defense ministries, the U.S.

Figure 3 - Total and Military Share of U.S. Integrated Circuit Sales



Source: Alic, John A., Lewis M. Branscomb, Harvey Brooks, Ashton B. Cater, and Gerald L. Epstein, « Beyond Spinoff: Military and Commercial Technologies in a Changing World » (Boston: Harvard Business School Press, 1992), Table 8-1, page 260.

military awarded substantial procurement contracts to new entrants such as Texas Instruments, with little or no history of supplying the military (2).

The U.S. Defense Department's willingness to purchase from untried suppliers was accompanied by policies that required suppliers to develop a « second source » for the product, *i.e.*, a domestic producer that could manufacture an electronically and functionally identical product, in order to avoid supply interruptions. Compliance with second source requirements meant that firms had to exchange design and process knowledge to ensure that the component produced by a second source was identical to the original product. These « second-source » policies thus resulted in considerable knowledge and technology transfer among suppliers, many of which were themselves recent entrants to the semiconductor industry. The knowledge sharing that resulted was an important factor in the high rates of new-firm formation in this industry, which in turn contributed to strong interfirm competition and innovation.

As nondefense demand grew to dominate the market for semiconductor components, defense-civilian technology « spillovers » declined in significance and actually reversed in direction. By the 1970s, « military specification » semiconductor components often lagged behind their commercial counterparts in technical performance, although these « milspec » components could operate in much more « hostile » environments of high temperatures or vibration. Concern among U.S. defense policymakers over this « technology gap » resulted in the creation of the DoD Very High Speed Integrated Circuit program (VHSIC) in 1980, which sought to advance military semiconductor technology more rapidly (See Stowsky, 1992, for one account). Originally planned for a six-year period and budgeted at slightly more than \$200 million, the VHSIC program lasted for 10 years and funded R&D and related contracts that amounted to nearly \$900 million. Nonetheless, the program failed to meet its objectives, demonstrating the limited influence of the federal government within a U.S. semiconductor market that by the 1980s was dominated by commercial applications and products.

Computer hardware

During World War II, the American military sponsored a number of projects to develop high-speed calculators for computing trajectories and firing tables

- (2) « European governments provided only limited funds to support the development of both electronic component and computer technology in the 1950s and were reluctant to purchase new and untried technology for use in their military and other systems. European governments also concentrated their limited support on defense-oriented engineering and electronics firms. The American practice was to support military technology projects undertaken by industrial and business equipment firms that were mainly interested in commercial markets. These firms viewed their military business as a development vehicle for technology that eventually would be adapted and sold in the open marketplace ». (Flamm, 1988, p. 134).

Figure 4 - Military-funded computer projects, 1945-55

First Generation of U.S. Computer Projects	Estimated Cost of Each Machine (thousands of dollars)	Source of Funding	Initial Operation
ENIAC	750	Army	1945
Harvard Mark II	840	Navy	1947
Eckert-Mauchly BINAC	178	Air Force (Northrop)	1949
Harvard Mark III	1,160	Navy	1949
NBS Interim computer (SEAC)	188 ^a	Air Force	1950
ERA 1101 (Atlas I)	500	Navy/NSA ^b	1950
Eckert-Mauchly UNIVAC	400-500	Army via census; Air Force	1951
MIT Whirlwind	4,000-5,000	Navy; Air Force	1951
Princeton IAS computer	650 ^a	Army; Navy; RCA; AEC	1951
Univ. of Cal. CALDIC	95 ^a	Navy	1951
Harvard Mark IV	n.a.	Air Force	1951
EDVAC	467	Army	1952
Raytheon Hurricane (RAYDAC)	460 ^a	Navy	1952
ORDVAC	600	Army	1952
NBS/UCLA Zephyr computer (SWAC)	400	Navy; Air Force	1952
ERA Logistics computer	350-650	Navy	1953
ERA 1102 (3 built)	1,400 ^c	Air Force	1953
ERA 1103 (Atlas II, 20 built)	895	Navy/NSA	1953
IBM Naval Ordnance Research Computer (NORC)	2,500	Navy	1955

Source: Flamm, « Creating the Computer », page 76

n.a. Not available

a. Estimated cost in 1950, in « Report on Electronic Digital Computers by the Consultants to the Chairmand of the Research and Development Board, » June 15, 1950, app. 4, cited by Kent C. Redmond and Thomas M. Smith, « Project Whirlwind: The History of a Pioneer Computer » (Digital Press, 1980), p. 166.

b. The National Security Agency (NSA) includes Army and Navy predecessor agencies.

c. Cost for three machines.

for artillery. The ENIAC, introduced in 1945 and generally considered the first fully electronic U.S. digital computer, was funded by Army Ordnance for this purpose, but military applications for computers rapidly expanded to include such areas as nuclear weapons design, cryptography, and strategic defense, and a number of military-funded computers were developed during the 1940s and 1950s (Figure 4). From the earliest days of their support for the development of computer technology, the U.S. armed forces supported wide diffusion of

technical information on this innovation. This attitude, which contrasted with that of the military in Great Britain or the Soviet Union, appears to have stemmed from the U.S. military's concern that a substantial industry and research infrastructure would be required for the development and exploitation of computer technology (3).

Even after the emergence of a substantial private industry dedicated to the development and manufacture of computer hardware, defense-related R&D spending aided the creation of the new academic discipline of computer science. Universities were important sites for applied, as well as basic, research in hardware and software, and contributed to the development of new hardware and networking technologies (see below for further discussion). The training by universities of engineers and scientists active in the computer industry also was important. U.S. universities provided important channels for cross-fertilization and information exchange in computer science research between industry and academia, but also between defense and civilian research efforts. Federal support for university research in the nascent discipline of computer science strengthened the research and knowledge network linking the federal government, academia, and industry.

As was true of semiconductors, federal R&D support was supplemented by procurement spending on military systems. Just as had been the case in semiconductors, the military market attracted new firms to enter the industry, and many such enterprises entered the fledgling U.S. computer industry in the late 1950s and 1960s.

Computer software

By the 1980s, the development of the semiconductor and computer industries had laid the groundwork for the expansion of another « new » postwar industry, the production of standardized computer software for commercial markets (as opposed to the production of custom software for internal use).

- (3) Goldstine, one of the leaders of the wartime project sponsored by the Army's Ballistics Research Laboratory at the University of Pennsylvania that resulted in the Eckert-Mauchly computer, notes that « A meeting was held in the fall of 1945 at the Ballistic Research Laboratory to consider the computing needs of that laboratory "in the light of its post-war research program". The minutes indicate a very great desire at this time on the part of the leaders there to make their work widely available. "It was accordingly proposed that as soon as the ENIAC was successfully working, its logical and operational characteristics be completely declassified and sufficient be given to the machine... that those who are interested... will be allowed to know all details" », (Goldstine, 1993, p. 217). Goldstine is quoting the « Minutes, Meeting on Computing Methods and Devices at Ballistic Research Laboratory », 15 October 1945 (note 14).

Procurement of products and services by the federal government was an important factor in the early development of the U.S. software industry. IBM was the primary supplier of computers for the first large-scale military computer program, the SAGE air defense project of the 1950s, but the RAND Corporation was the contractor responsible for the bulk of the huge amount of software required for SAGE. The SAGE software development project acted as a « university » of sorts for hundreds software programmers, laying the foundations for the software industry's future development within the United States (Campbell-Kelly, 1995).

In the late 1950s and early 1960s, defense contractors, including TRW, MITRE Corporation, and Hughes, began to produce large-scale systems software for military applications under federal contracts. IBM and other mainframe computer manufacturers also produced large custom software applications for customers and became important suppliers in the software-contracting industry. Much of the software-related knowhow developed from defense contracts « spilled over » to commercial applications, such as early airline reservation systems (Campbell-Kelly, 1995).

Like hardware, the emergent U.S. computer software industry benefited from postwar defense-related spending on computer science R&D in universities and industry. Through the 1970s and 1980s, almost half of the federal funds for mathematics and computer science research (which included funding from major civilian research agencies, such as the National Science Foundation and the National Institute of Health) went to universities and colleges. The defense share of federal computer science R&D funding declined from almost 60 % in fiscal 1986 to less than 30 % in fiscal 1990 (Clement, 1987, 1989; Clement and Edgar, 1988), and defense funding of computer science R&D in universities was supplanted by the growth in funding for quasi-academic research and training organizations.

Federal procurement programs influenced the evolution of specific programming languages as well. A DoD effort to establish a standard programming language resulted in the widely used « common business-oriented language », COBOL. The DoD required that computers purchased by the military support COBOL, and that any business-related applications for defense programs be written in the language. Since DoD accounted for such a large share of the « market » for custom software, its procurement requirements facilitated the development and diffusion of COBOL (Flamm 1987, p. 76).

Much of the rapid growth in custom software firms during the 1969-80 period reflected expansion in federal demand, which in turn was dominated by DoD demand. There exists no reliable time series of DoD expenditures on software procurement that employs a consistent definition of software. Nevertheless, the available data suggest that in constant-dollar terms, DoD expenditures on software increased more than thirtyfold during the 1964-90 period (Langlois and Mowery, 1996). Throughout this period, DoD software demand was dominated by custom software, and DoD and federal government

markets for custom software accounted for a substantial share of the total revenues in this segment of the U.S. software industry. Procurement spending promoted learning, knowledge development, and standards in the early stages of the industry. Just as occurred in computer hardware and semiconductors, however, commercial demand grew to outstrip defense-related markets. By the early 1990s, defense demand accounted for a declining share of the U.S. software industry's revenues.

The tangled history of the Defense Department's « generic » software language, Ada, unveiled in 1984, illustrates the declining influence of federal procurement on the rapidly growing software industry. Billed as a solution to the problems of system maintenance and software development resulting from the bewildering variety of software languages in use within defense systems, Ada was designed as a single « standard » software language for all defense applications.

Ada proponents argued that the creation of a software standard for military applications would attract commercial developers, many of whom no longer pursued opportunities in the defense market, to produce software that could be used in both civilian and military applications. But the difficulties associated with « inserting » Ada into the enormous « installed base » of defense-related software meant that the language failed to attract the attention of commercial developers. The contrast between the failure of this DoD-supported language to take hold and COBOL's rapid diffusion into military and commercial applications underscores the points made earlier concerning the tendency for the influence of defense-related R&D and procurement demand to decline as commercial markets expand.

The Internet

The Internet was invented and commercialized primarily in the United States, although scientists and engineering in other industrial economies (especially France and the United Kingdom) made important contributions to computer networking technologies during the 1970s, and the key technologies underpinning the « World Wide Web » were invented at CERN, the European nuclear physics research facility. Nonetheless, U.S. entrepreneurs and firms led the transformation of these inventions into components of a national and global network or networks, and were early adopters of new applications.

The Department of Defense played a critical role in funding the development and diffusion of early versions of the technology. Federal spending on R&D and procurement was complemented by the R&D investments of large corporations and the many start-ups that populated Internet-related industries. During the early 1960s, several researchers, including Leonard Kleinrock at MIT and Paul Baran of RAND, developed various aspects of the theory of packet switching (4). The work of Baran, Kleinrock and others led the U.S.

Department of Defense Advanced Research Projects Agency (DARPA) to fund the construction of a prototype network. In December 1968, DARPA granted a contract to the Cambridge Massachusetts-based engineering firm of Bolt, Beranek and Newman (BBN) (5) to build the packet switches that linked computers at several major research computing facilities. The resulting ARPANET is widely recognized as the earliest forerunner of the Internet (NRC, 1999, Ch. 7). By 1975, as universities and other major defense research sites were linked to the network, ARPANET had grown to more than 100 nodes.

In 1974, two DARPA-funded engineers, Robert Kahn and Vinton Cerf, published the first version of the TCP/IP protocol suite (6). The new data-networking protocol allowed physically distinct networks to interconnect with one another as « peers » and exchange packets through special hardware, called a gateway. By publishing TCP/IP and placing the standard within the public domain, Kahn, Cerf and their collaborators strengthened its position in a two-decade-long competition among a variety of networking protocols, including proprietary standards such as IBM's SNA and Digital equipment's DECNET, open alternatives such as Datagram (UDP) networking and the Unix to Unix Copy protocol (UUCP), and standards supported by established telecommunications firms.

TCP/IP's origins in a federally funded research project (and its development at a time during which the results of federally funded research rarely were patented) was crucial to the eventual « victory » of this open, nonproprietary standard as the foundation for the architecture of the Internet. Had a proprietary protocol become dominant, the evolution of computer networking and eventually, the Internet, might have been very different (See Mowery and Simcoe, 2002, for further discussion). The resulting widespread diffusion of the Internet's core technological innovations lowered barriers to the entry by networking firms in hardware, software and services.

- (4) On a packet-switched network, information is broken up into a series of discrete « packets » that are sent individually, and reassembled into a complete message on the receiving end. A single circuit may carry packets from multiple connections, and the packets for a single communication may take different routes from source to destination.
- (5) Bolt, Beranek and Newman, an MIT « spinoff » founded in 1948, was an early example of the new firms that played an important role in the Internet's development. The firm was started by MIT Professors Bruce Bolt and Leo Beranek in partnership with a graduate student, Robert Newman.
- (6) « Transmission Control Protocol (TCP) » in the *IEEE Transactions on Communication*, (also RFC 0675, 1974).

U.S. dominance in computer networking did not result from a first-mover advantage in the invention or even the early development of a packet-switched network. French and British computer scientists also contributed important technical advances to packet-switching and computer-networking technologies and protocols during this period, and publicly supported prototype computer networks were established in both France and the UK by the early 1970s. But the ARPANET differed from the French and British projects in its funding for large-scale deployment of the network. Its size and inclusion of a diverse array of institutions as members both appear to distinguish the ARPANET from its British and French counterparts, and accelerated the development of supporting technologies and applications.

The large scale of the U.S. defense-related programs in computer science research and networking distinguished them from similar programs in the United Kingdom and France. In addition to their size, the structure of these substantial federal R&D investments enhanced their effectiveness. In its efforts to encourage exploration of a variety of technical approaches to research priorities, DARPA frequently funded similar projects in several different universities and private R&D laboratories. Moreover, the Department of Defense's procurement policy complemented DARPA's broad-based approach to R&D funding. As had been true of semiconductors, the award of major development and procurement contracts to small firms such as BBN helped foster entry by new firms into the emerging Internet industry, supporting intense competition and rapid innovation.

In May 1991, Tim Berners-Lee and Robert Cailliau, two physicists working at the CERN laboratory in Switzerland, released a new document format called Hyper-Text Markup Language (HTML) and an accompanying document retrieval protocol called Hyper-Text Transfer Protocol (HTTP) (7). HTML and HTTP were not invented in the United States, but decades of previous federal investments in R&D and infrastructure supported their rapid domestic adoption and development. Although it was not the first functional Internet « browser », Mosaic, a free program written by a group of graduate students at the University of Illinois' National Center for Supercomputing Applications, was widely adopted and accelerated the growth of the Web (8). During 1993, the

- (7) The development of these technical advances was motivated by Berners-Lee and Cailliau's interest in facilitating the ability of physicists to archive and search the large volumes of technical papers being transmitted over the Internet as it then existed.
- (8) NCSA was an NSF-funded facility devoted to research on supercomputing architecture and applications. By the early 1990s, networking technologies and powerful desktop computers had reduced the need of academic researchers for access to supercomputers. As a result, NCSA researchers focused on developing new technologies to support expanded use of computer networking (Abbate 1999, p. 216).

first year that Mosaic was available, HTTP traffic on the Internet grew by a factor of 3,416. By 1996, HTTP traffic was generating more packets than any other Internet application.

CONCLUSION

The postwar U.S. economy has benefited from the creation and innovative dynamism of a series of « new industries », notably in information technology. It is difficult to overstate the importance of defense-related support for R&D and, especially in industries such as computer hardware or semiconductors, procurement, which together accelerated the growth of U.S. firms. The structure of these public programs also enhanced the innovative dynamism and competitive strength of the IT sector. But the influence of defense-related funding for R&D and procurement was enhanced by the fact that such public intervention occurred at the inception of these industries. As the IT sector developed, DoD R&D and procurement contracts exercised less influence over the semiconductor, computer hardware, and computer software industries in the United States.

The sheer scale of U.S. defense-related R&D and procurement spending distinguishes the postwar U.S. defense establishment from those of other OECD economies. Other important differences, such as the willingness of at least some U.S. defense program managers to place substantial orders for new systems and components with young firms, as well as the prominent role of universities as performers of a considerable amount of DoD-funded research, also distinguish the U.S. programs from those of such nations as Japan, France, or the United Kingdom. These policies contributed to the inter-institutional linkages emphasized by Freeman (1995), as well as the creation of an R&D « infrastructure » that Cowan and Foray (1995) discuss as an important source of economic benefits from military R&D and procurement. Nevertheless, many of these « structural components » were themselves made feasible by the sheer scale of U.S. defense-related programs.

The economic and institutional effects of U.S. defense-related R&D spending in the wartime period of the 1940s and the postwar Cold War also were influenced by the fact that a large share of these federal R&D investments were channeled to private firms and extramural research performers, rather than the governmental arsenals that accounted for a large share of pre-1940 defense-related R&D in the United States. In addition, of course, the period of hot and cold hostilities in which the United States was engaged after 1940 extended over a far longer period than was true of these earlier conflicts. The global « war on terror » declared by the current U.S. Administration portends a renewed technological and economic mobilization that may last for decades.

Will the post-9/11 resurgence in defense and « homeland security » spending produce large federal investments in IT-related R&D? Figure 1 reveals the reversal of a decades-long decline in the defense-related share of federal R&D

spending and increase in nondefense federal R&D spending in the federal budgets revised or prepared in the aftermath of the 9/11 terrorist attacks on the U.S. homeland. A similar and simultaneous reversal in the longstanding decline in the federal-government share of total national R&D spending is apparent in Figure 2. These trends reflect the development of a new and ambitious R&D agenda in counterterrorism, especially in the development of technologies to counter chemical, biological, and radioactive weapons, as well as the increases in overall defense spending that followed the U.S. invasions of Afghanistan and Iraq. Any assessment of the long-term effects of a dramatic shift in federal spending priorities that is itself less than 4 years old must be very provisional. But several elements of post-9/11 federal R&D spending trends, if not all of their long-term effects, are noteworthy.

DoD-funded « research » spending increased in real-dollar terms during fiscal 2001-2005, although its share of overall defense-related R&D spending has declined. But the fiscal 2006 budget released by the White House in February 2005 reduces the amount budgeted for defense-related research by more than 20 % from the fiscal 2005 level approved by Congress (All budgetary data are taken from American Association for the Advancement of Science, 2005). It is likely that the Congress will increase the amount allocated to this area, but the overall trend since 9/11 is unmistakable—defense-related R&D investment is shifting toward a shorter time horizon and a budget that is dominated to an even greater extent by development work. Such a shift is likely to reduce the « pure knowledge spillovers » described above, and could slow growth in or actually reduce defense-related investments in academic research in engineering and the physical sciences.

Still another new feature of the post-9/11 shifts in U.S. defense-related R&D spending is spending on R&D related to « homeland security ». Trends in « homeland security » R&D spending for fiscal 2002-2006 reveal two interesting characteristics. The first and unsurprising feature of the homeland security R&D budget is its rapid growth, increasing in constant-dollar terms from roughly \$ 1.5 billion in fiscal 2002 (an amount that reflects increases made in that year's budget after 9/11) to \$ 4.4 billion in fiscal 2006 (the FY 2006 amount is the request in the Presidential budget issued in February 2005, and is likely to be modified by Congress). The second and perhaps more surprising aspect of the homeland security R&D budget is the extent to which it is dominated by biomedical R&D, which is included in the spending associated with the Department of Health and Human Services (HHS, the federal agency that oversees both the National Institutes of Health and the Centers for Disease Control and Prevention). Indeed, R&D spending related to homeland security associated with HHS (\$1.8 billion requested in the FY2006 budget) outstrips the spending allocated to the federal Department of Homeland Security (\$1.3 billion). Although some of the homeland security R&D budget will go to information technology, this component of increased federal spending on national security-related R&D (as the term has been redefined since 9/11) seems likely to favor biomedical R&D.

Throughout the post-1945 period, the influence of federal R&D and procurement on the technological and structural development of the IT sector was amplified by other policies in such areas as antitrust policy and intellectual property rights. Remarkably, in light of the consistently pro-competitive posture of all of these disparate policies during the 1945-70 period, there is little evidence of systematic coordination among them. Some of the beneficial effects of these DoD R&D and procurement programs were rooted in their encouragement for relatively liberal disclosure and interfirm diffusion of technology. This posture contrasts with today's conventional wisdom about the desirability of patenting the intellectual property resulting from federal R&D, conventional wisdom now embodied in such policies as the Bayh-Dole Act of 1980 (See Mowery *et al.*, 2001). Although the « evidence » provided by these histories of selected high-technology industries must be treated with caution, they suggest that relatively weak intellectual property rights regimes may prove beneficial, or at least not harmful, to the development and innovative performance of industries in their early stages of development.

REFERENCES

- ABBATE, J., « Inventing the Internet » (Cambridge : MIT Press, 1999).
- ALIC John A., Lewis M. BRANSCOMB, Harvey BROOKS, ASHTON B. CATER, and Gerald L. EPSTEIN, « Beyond Spinoff: Military and Commercial Technologies in a Changing World » (Boston : Harvard Business School Press, 1992).
- American Association for the Advancement of Science, « Research and Development in the FY2003 Budget » (Washington, D.C. : American Association for the Advancement of Science, 2002).
- American Association for the Advancement of Science, « Research and Development in the FY2006 Budget » (Washington, D.C. : American Association for the Advancement of Science, 2005).
- BERNERS-LEE T., « Weaving the Web » (New York : HarperCollins, 2000).
- BEST M.-H., and R. FORRANT, « Creating Industrial Capacity: Pentagon-Led versus Production-Led Industrial Policies », in J. Michie and J.G. Smith, eds., « Creating Industrial Capacity » (Oxford : Oxford University Press, 1996).
- BORRUS M., « Competing for Control » (Boston : Ballinger, 1988).
- CAMPBELL-KELLY M., « Development and Structure of the International Software Industry, 1950-1990 », *Business and Economic History* 24, 1995, 73-110.
- CERF V. *et al.*, « A Brief History of the Internet », <www.isoc.org/internet/history/>.
- CLEMENT J.R.B., « Computer Science and Engineering Support in the FY 1988 Budget », in Intersociety Working Group, ed., AAAS Report XII: Research & Development, FY 1988 (Washington, D.C. : American Association for the Advancement of Science, 1987).
- CLEMENT J.R.B., « Computer Science and Engineering Support in the FY 1990 Budget », in Intersociety Working Group, ed., AAAS Report XIV: Research & Development, FY 1990 (Washington, D.C. : American Association for the Advancement of Science, 1989).
- CLEMENT J.R.B. and D. EDGAR, « Computer Science and Engineering Support in the FY 1989 Budget », in Intersociety Working Group, ed., AAAS Report XIII: Research & Development, FY 1989 (Washington, D.C. : American Association for the Advancement of Science, 1988).

- COWAN R., and D. FORAY, « Quandaries in the economics of dual technologies and spillovers from military to civilian research and development », *Research Policy* 24, 1995, 851-868.
- DERTOUZOS M., R. LESTER, and R. SOLOW, « Made in America » (Cambridge, MA : MIT Press, 1989).
- FLAMM Kenneth, « Targeting the Computer » (Washington, DC : The Brookings Institution, 1987).
- FLAMM Kenneth, « Creating the Computer » (Washington, DC : The Brookings Institution, 1988).
- FLORIDA R., and M. KENNEY, « The Breakthrough Illusion » (New York : Basic Books, 1990).
- FREEMAN C., « The "National System of Innovation" in Historical Perspective », *Cambridge Journal of Economics* 19, 1995, 5-24.
- GOLDSTINE H.H., « The Computer from Pascal to Von Neumann », 2nd ed. (Princeton, NJ : Princeton University Press, 1993).
- KLEINMAN H., « The Integrated Circuit Industry : A Case Study of Product Innovation in the Electronics Industry », unpublished D.B.A. dissertation, George Washington University, 1966.
- LANGLOIS R. N., and D. C. MOWERY, « The federal government role in the development of the U.S. software industry » in D. C. MOWERY ed., « The International Computer Software Industry : A Comparative Study of Industry Evolution and Structure » (New York : Oxford University Press, 1996).
- LESLIE S.W., « The Cold War and American Science » (New York : Columbia University Press, 1993).
- LEVIN RICHARD C., « The Semiconductor Industry », in RICHARD R. NELSON ed., « Government and Technical Progress : A Cross-Industry Analysis » (New York : Pergamon Press, 1982).
- LICHTENBERG F., « The relationship between federal contract R&D and company R&D », *American Economic Review* 74, 1984, 73-78.
- LOWEN R.S., « Creating the Cold War University » (Berkeley, CA : University of California Press, 1997).
- MOWERY D.C., and T. SIMCOE, « The History and Evolution of the Internet », in B. STEIL, R. NELSON, and D. VICTOR, eds., « Technological Innovation and Economic Performance » (Princeton, NJ : Princeton University Press, 2002).
- MOWERY D.C., R.R. NELSON, B.N. SAMPAT, and A.A. ZIEDONIS, « The Growth of Patenting and Licensing by U.S. Universities : An Assessment of the Effects of the Bayh-Dole Act of 1980 », *Research Policy* 30, 99-119, 2001.
- MOWERY D.C., and N. ROSENBERG, « Paths of Innovation » (New York : Cambridge University Press, 1999).
- National Research Council, « Funding a Revolution : Government Support for Computing Research », (National Academy Press, 1999).
- REDMOND Kent C., and Thomas M. SMITH, « Project Whirlwind : History of a Pioneer Computer » (Bedford, MA : Digital Press, 1980).
- SAMUELS R.J., « Rich Nation, Strong Army : National Security and the Technological Transformation of Japan » (Ithaca, NY : Cornell University Press, 1994).
- SCHANKERMAN M., and A. PAKES, « Estimates of the value of patent rights in European countries during the post-1950 period », *Economic Journal* 96, 1986, 1052-1076.
- STOWSKY J., « From Spin-off to Spin-on : Redefining the Military's Role in American Technology Development », in W. SANDHOLTZ, M. BORRUS, J. ZYSMAN, K. CONCA, J. STOWSKY, S. VOGEL, and S. WEBER, « The Highest Stakes » (New York : Oxford University Press, 1992).
- TANSEY B., « An Insider's View of Nanotechnology », *San Francisco Chronicle*, February 1, 2004.
- YUDKEN J.S., and B. SIMONS, « Computer Science Research Funding : Issues and Trends », *Abacus*, 1988, 60-66.